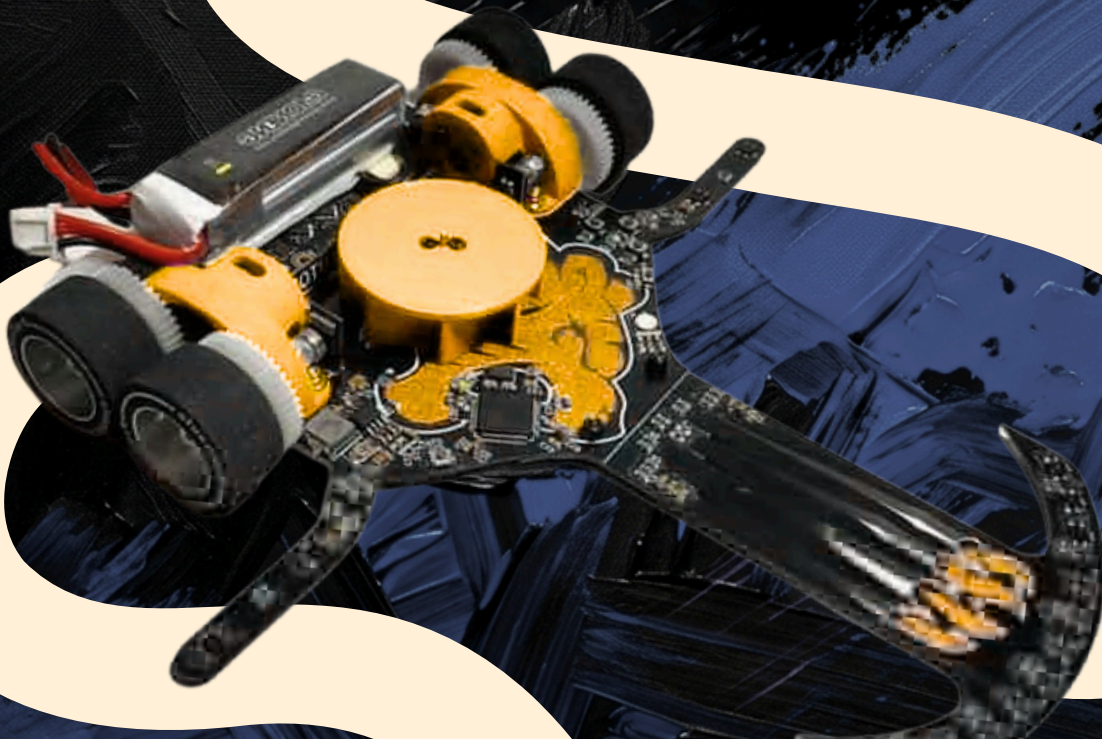


OFFICIAL RULEBOOK - 2026

KYNEX

LINE FOLLOWER 26TH SEP 2026

One track, one line, zero margin for error. Build an autonomous racer that reads the path faster and cleaner than anyone else. Read every rule before you build





KYNEX

BEFORE YOU REGISTER TEAM & ELIGIBILITY

4

MEMBERS

MAX TEAM SIZE

1

ROBOT

PER TEAM

2

MEMBERS

ALLOWED AT TRACKSIDE

TEAM COMPOSITION (0.1)

- A team may have a **maximum of 4 members** on its roster, and may field **one bot only**.
- During a match, only **2 team members** are permitted at the track side - the rest must remain in the designated team / spectator area.
- The bot must be built so that **only one person** operates it at any given time.
- Only one member is designated as the operator for each run.

ELIGIBILITY (0.2)

- **All undergraduate students** are eligible to participate.
- Participants must carry a **valid institutional ID card** at all times during the event.
- Every bot must clear inspection before it is allowed to compete.

CODE OF CONDUCT (0.3)

- **General Behaviour:** All participants must treat fellow competitors, judges, volunteers, and organizers with respect at all times. Unsportsmanlike conduct, harassment, intimidation, or discriminatory behavior of any kind will result in immediate disqualification. Teams are responsible for the conduct of all their members and supporters.
- **Competition Integrity:** Any attempt to sabotage another team's robot or interfere with another team's run is strictly prohibited. Sharing real-time track information between competing teams during active rounds is not permitted. Any form of cheating or deliberate rule exploitation will result in disqualification and may result in a ban from future KYNEX events.
- **Equipment & Environment:** Teams must not tamper with the track, lighting, or any official competition equipment. Robots must not be modified in any way once a competition round has officially begun.
- **Media & Photography:** Participants may photograph or film for personal and team use unless otherwise instructed by organizers. Official KYNEX media personnel take priority in designated filming areas.



KYNEX

KNOW THE BASICS DEFINITIONS

WHY THIS SECTION EXISTS (1.0)

Every term below is used with a practice meaning through the rest of this rulebook. When in doubt, this section is the reference.

● RUN (1.1)

A single official attempt by a robot to complete the track.

● LINE LOSS (1.2)

Failure to maintain valid line tracking for more than 3 consecutive seconds.

● VALID PATH (1.3)

The correct route from start to finish as defined on the official track map.

● FORWARD PROGRESS (1.4)

Movement toward the finish as defined on the official track map.

● RESET (1.5)

The act of returning a robot to the start line, or to a designated recovery point, under judge supervision after a run has been paused. A reset is not a rerun; time already elapsed is not restored unless the judge rules otherwise under section 5.3.

● RERUN (1.6)

A complete, fresh attempt at the track, granted only under the conditions in section 5.3, with the run clock and penalty count reset to zero.

● DISQUALIFICATION (1.7)

Removal of a robot from a single run, a full match, or the entire event, per the severity levels defined in section 6.3. A DQ from a single run does not by itself remove the team from the rest of the competition unless the violation falls under "Serious" or "Extreme" in section 6.3.



KYNEX

BUILD CONSTRAINTS

HARDWARE RULES & BOT SPECIFICATIONS

250 x 250
MM X MM
MAX FOOTPRINT

250
MM
MAX HEIGHT

1
KG
MAX WEIGHT

AUTONOMY (2.1)

The robot must operate fully autonomously throughout the entire run. Any form of remote control, wireless communication (Wi-Fi, Bluetooth, RF or tethered control) during a run, or external intervention is strictly prohibited. Violation of this rule results in immediate disqualification from that run.

PERMITTED CONTROLLERS (2.2)

- Arduino-based boards (Uno, Mega, Nano, etc.)
- ESP32 / ESP8266
- Custom PCBs (must pass technical inspection)
- STM32 micro-controllers

RESTRICTIONS (2.3)

PROHIBITED: Pre-built line-follower kits, LEGO-based robots and LEGO Mindstorms controllers are not permitted under any circumstances.

- All sensor types are permitted without restriction (IR, ultrasonic, etc.).
- Robots must be battery powered only no wired external power. Maximum voltage: 12 V DC.

SAFETY (2.4)

A physical emergency stop button is mandatory and must immediately disable all motion. All components must be securely mounted and must not detach during operation. Robots deemed unsafe by the inspection team will be disqualified.

MODIFICATIONS & CALIBRATION (2.5)

No modifications are permitted during a run. Adjustments are only permitted before the run officially begins, during the calibration window defined in section 4.6.

TECHNICAL INSPECTION (2.6)

All robots must pass a mandatory technical inspection before competing. Inspection verifies dimensions, weight, safety compliance, and controller legality. If a robot fails inspection, it may request re-inspection at the judges' discretion, within the available inspection window. Robots that do not pass inspection before the competition begins may not participate.



KYNEX

KNOW YOUR BATTLEFIELD

TRACK SPECIFICATIONS

ABOUT THE TRACK (3.1)

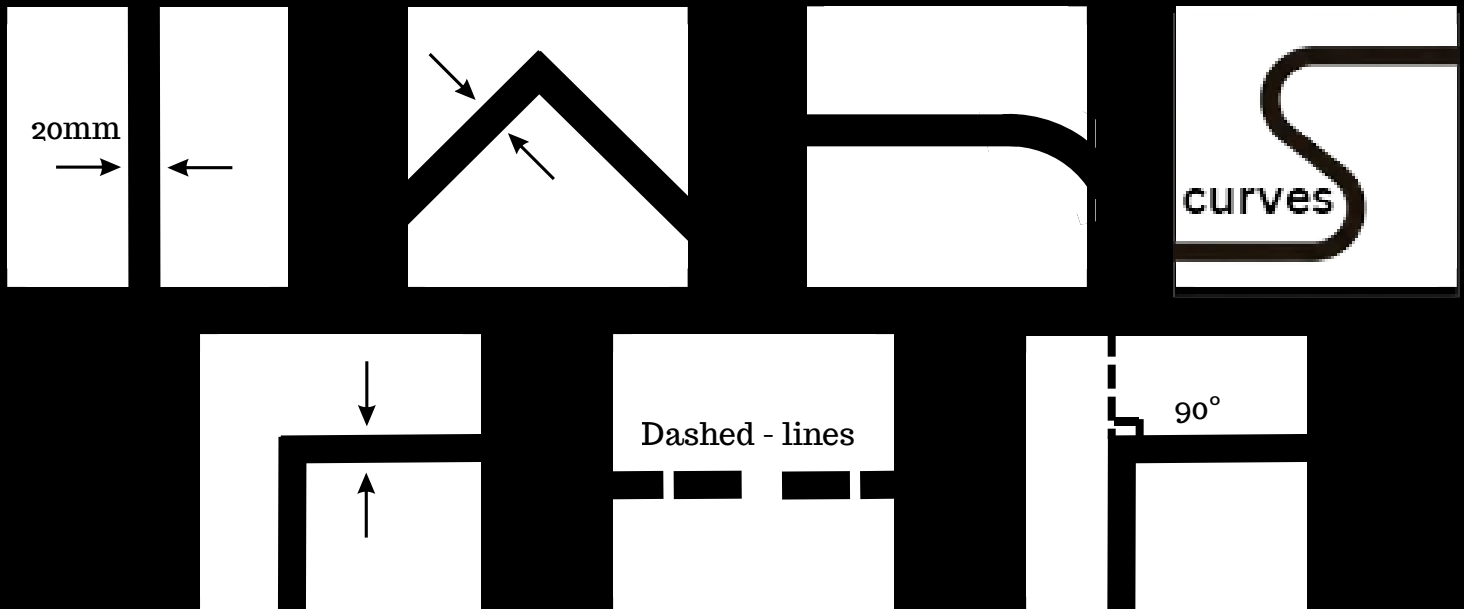
Dimensions: Main Arena dimensions are 2.5m x 2.5m

Surface: White matte, non-reflective surface. The layout of the track shall remain strictly confidential until the day of the competition. The track will be revealed to all teams simultaneously prior to commencement.

Line: Black line, constant width of 2 cm $\pm$ 2%.

TRACK ELEMENTS (3.2)

- Sharp Turns — up to 90°
- Zigzags — multiple consecutive direction changes
- Curves — gradual bends along the path
- Dashed Lines — non-continuous segments of the line
- Intersections — path crossings requiring correct routing



DASHED LINE SPECIFICATIONS (3.3)

Dashed segments consist of black marks no shorter than 3 cm, separated by gaps no wider than 5 cm, so that a compliant sensor array can bridge each gap. Gap width is subject to the $\pm$ 5% tolerance in section 3.4.

TOLERANCE (3.4)

A $\pm$ 5% dimensional variation is permitted across all track elements.



KYNEX

HOW YOU WIN SCORING SYSTEM

SCORING MODEL & RANKING CRITERIA (4.1)

KYNEX '26 uses a performance-time scoring model: robots are scored primarily on whether they finish, then on how quickly they do it.

Final ranking is determined in this strict priority order:

1. Completion — any robot that crosses the finish line under forward motion (section 4.7) ranks above any robot that does not.
2. Time — among robots with the same completion result, the lower Final Score (section 4.8) ranks higher.

TIE-BREAKER PROCEDURE (4.2)

If two or more robots are tied on both Completion and Final Score (to the nearest 0.01 s), ranking is decided in this order: (a) fewer touch penalties under section 4.4; (b) greater Forward Progress if neither completed.

NO. OF RUNS PER TEAM (4.3)

Each team receives one official Run per round, unless otherwise communicated to teams at the venue on the day of the competition, where the total number of permitted Runs will be announced. Additional attempts are granted only as Reruns under section 5.3.

PENALTIES (4.4)

Whenever a team member touches the robot during a run other than to press the emergency stop or during a judge-supervised Reset a penalty of 5 seconds is added to that team's total time for the run. Repeated or extended contact (e.g., picking up or repositioning the robot) is handled under section 4.5 rather than this penalty.

DISQUALIFYING ACTIONS (4.5)

A run (not the team's entire event) is disqualified if the robot: is remotely controlled or receives wireless assistance (section 2.1); is physically carried, lifted, or repositioned by a team member outside of an official Reset; exploits reverse motion for strategic gain rather than line recovery (section 5.1); or is modified during the run (section 2.5).

TIME LIMIT (4.6)

Each run has a maximum duration of 3 minutes (180 seconds). Each team is allotted 45 seconds for calibration immediately before its run. The run ends when the robot crosses the finish line or time expires.

COMPLETION (4.7)

The robot must fully cross the finish line using forward motion only. Completion is the highest-priority ranking factor.

FINAL SCORE FORMULA (4.8)

Final Score (seconds) = Raw Run Time + (5 s × number of touch penalties)

Raw Run Time is measured from the start signal to the moment the robot fully crosses the finish line. A robot that does not complete the run within the 180-second limit receives no Final Score and is ranked below all robots that completed, per section 4.1 and section 4.2.



KYNEX

STEP INTO THE ARENA MATCH FORMAT

◆ GAME FLOW (5.1)

- Each match consists of one official Run per team, in the order set by the published schedule, followed by any Reruns granted under section 5.3. See section 4.3 for the number of runs per team.
- The robot must be fully behind the start line before the signal. No movement is permitted before the official start signal. Team readiness must be confirmed verbally to the judge before the run begins.
- No team member may handle or touch the robot after it has been placed on the track, except to press the emergency stop or during a judge-supervised Reset. Any other contact is penalized under section 4.4; contact that amounts to carrying or repositioning the robot disqualifies the run under section 4.5. No external assistance of any kind is permitted.
- The run ends when the robot crosses the finish line.
The run ends when the 3-minute time limit expires.
The run ends if terminated by a judge (see section 5.4).
- Forward motion is the primary direction of travel. Reverse movement is permitted only during line recovery — exploitation of reverse motion for strategic gain is prohibited and is a section 4.5 disqualifying action.

LINE RECOVERY PROCEDURE (5.2)

If a robot loses the line (section 1.2), it may use reverse or lateral movement to reacquire it without penalty, provided this happens autonomously and within the 3-minute time limit. There is no separate penalty for a Line Loss beyond the time it costs; a robot that cannot reacquire the line simply runs out of time under section 5.1.

RERUNS (5.3)

A rerun may be granted only in the event of a verifiable judge error or external interference not caused by the team. Maximum of 1 rerun per match. A granted rerun resets the run clock and penalty count to zero; it does not average with the original attempt.

RUN TERMINATION (5.4)

A judge may terminate a run if:

- The robot poses a safety risk.
- The robot becomes non-functional.

A terminated run is scored as a non-completion under section 3.2 unless the judge attributes the termination to a cause covered by section 5.3.



KYNEX

PLAY FAIR

JUDGING & GENERAL RULES

THE JUDGE HAS FINAL AUTHORITY OVER ALL RULINGS.

APPEAL PROCESS (6.1)

Appeals must be submitted verbally to the Judge within 5 minutes of the disputed ruling. Only the designated Team Leader may submit an appeal. Appeals must reference a specific rule by section number. All appeal decisions by the Judge are final.

An appeal may be rejected without further review if it is submitted after the 5-minute window, submitted by anyone other than the Team Leader, or does not cite a specific section of this rulebook.

GENERAL COMPETITION RULES (6.2)

- No external assistance is permitted to the robot during a run.
- No team member may handle or touch the robot after it has been placed on the track, except during an official Reset or to press the emergency stop see section 4.4 and section 5.1 for how other contact is handled.
- No interference between competing robots or teams is permitted.
- No modification of the environment (track, lighting, surface) is permitted.
- The official track is the final reference teams must adapt their robots to actual conditions.
- Rule exploitation is strictly prohibited; the spirit of the rules takes precedence.
- Organizers reserve the right to update or clarify rules before the event. All updates will be published on our website.

VIOLATION LEVEL (6.3)

- Minor misconduct (first offense)
- Repeated misconduct
- Serious misconduct (e.g., sabotage, cheating)
- Extreme violation (e.g., physical aggression)

CONSEQUENCE (6.4)

- Verbal Warning
- Run disqualification
- Full event disqualification
- Permanent ban from KYNEX events



KYNEX

MARK YOUR CALENDER VENUE & LOGISTICS

LOCATION

KJ Somaiya Vidyavihar university,
Vidyavihar, Mumbai – 400077

26TH SEP

FINAL REMINDERS

- Arrive early for inspection — bots that fail size, weight, or weapon checks will not be permitted to compete.
- Be punctual — teams arriving more than 10 minutes late face serious action, with all final decisions left to event officials.
- Keep your battery pack accessible; it may need to be detached and presented at inspection.
- Review every rule in this document before your final build — decisions made by event officials on match day are final.

Note: Organizers reserve the right to revise or clarify these rules ahead of the event. Any changes will be communicated to all registered teams before match day.

ANY DOUBTS?

F&Q Form:

<https://docs.google.com/forms/d/e/1FAIpQLSddxqW865aIJ2p1sqtNFyYtydLoE1eNgPIPPJMYvEolvlijQg/viewform?usp=header>

For any queries contact:

Paarth Mehta - +91 8169524405

Arpita Bhalekar - +91 9321127041



kjsse_nexusrobotics



nexusrobotics.kjsse@somaiya.edu